Problem 11.29

On January 1, 1980, Jack deposited 1,000 into Bank X to earn interest at the rate of j per annum compounded semiannually. On January 1, 1985, he transferred his account to Bank Y to earn interest at the rate of k per annum compounded quarterly. On January 1, 1988, the balance in Bank Y is 1,990.76. If Jack could have earned interest at the rate of k per annum compounded quarterly from January 1, 1980 through January 1, 1988, his balance would have been 2,203.76. Calculate the ratio k/j.

Solution

First use the hypothetical balance, which earns interest at rate k for eight years, or 32 quarterly periods:

$$\begin{aligned} 1{,}000\left(1+\frac{k}{4}\right)^{32} &= 2{,}203.76, \\ k &= 4\left((2.20376)^{1/32}-1\right) \\ &\approx 0.10. \end{aligned}$$

In the actual investment, the deposit earns interest at rate j for five years, or 10 semiannual periods, followed by interest at rate k for three years, or 12 quarterly periods. Thus,

$$\begin{aligned} 1{,}000\left(1+\frac{j}{2}\right)^{10}\left(1+\frac{k}{4}\right)^{12} &= 1{,}990.76, \\ j &= 2\left[\left(\frac{1.99076}{(1+k/4)^{12}}\right)^{1/10}-1\right] \\ &\approx 0.08. \end{aligned}$$

Therefore,

$$\boxed{\frac{k}{j} \approx \frac{0.10}{0.08} = 1.25}.$$

The rates are approximate because the account balances in the problem are given to the nearest cent.